Periodic Functions

Introduction
Identifying the Period of a Function Given Its Graph
Drawing the Graph of a Periodic Function
Inferring Values of a Periodic Function Given Its Graph

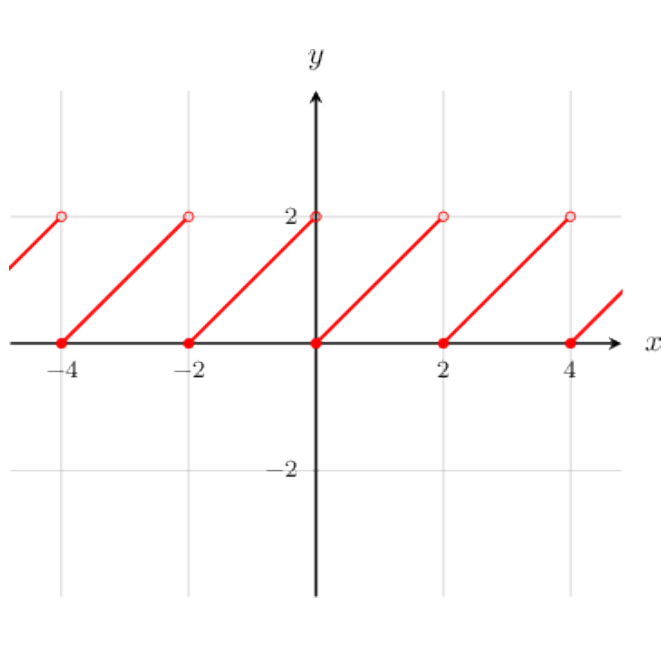
Prerequisites

Piecewise Functions

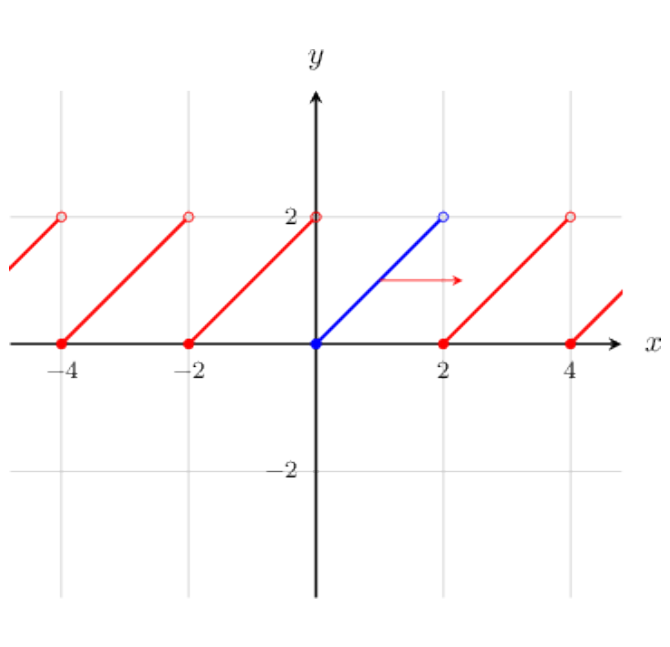
Introduction

Periodic functions are special functions that repeat themselves every c units. The smallest positive value of c that has this property is called the **period** of the function.

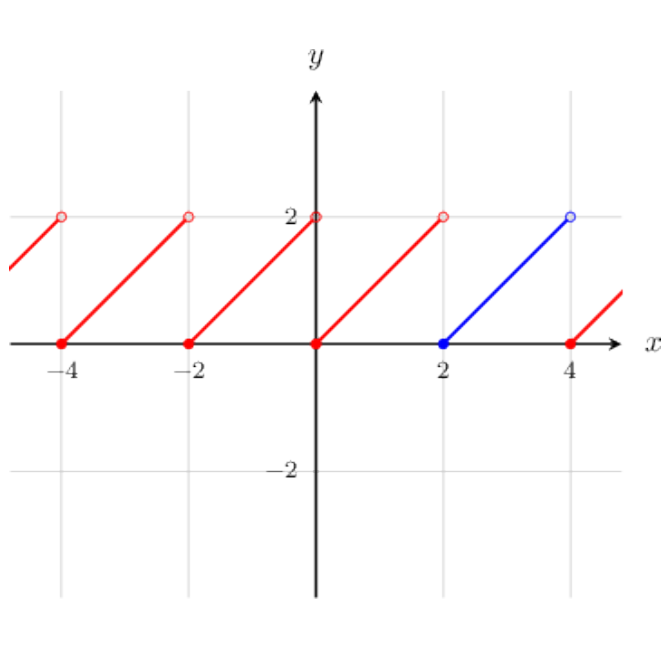
Let's take the following periodic function as an example:



This function repeats itself with a period of 2. To illustrate, first focus on the part of the graph shown in blue below.



If we shift this part 2 units to the right, the graph remains unchanged!



The same would happen if we shifted the part 2 units to the left.

By knowing the period of a periodic function, we can use a value of the function to infer many other values of the function. For example, in the function above, note that $f(4) = 0$. Since the period is 2, if we add (or subtract) any multiple of 2 from the input, the output stays the same. We can write this as

$$f(4 + 2n) = 0,$$

where n is an integer.

For example, setting $n = 1, 2, 3 \ldots$ gives the following:

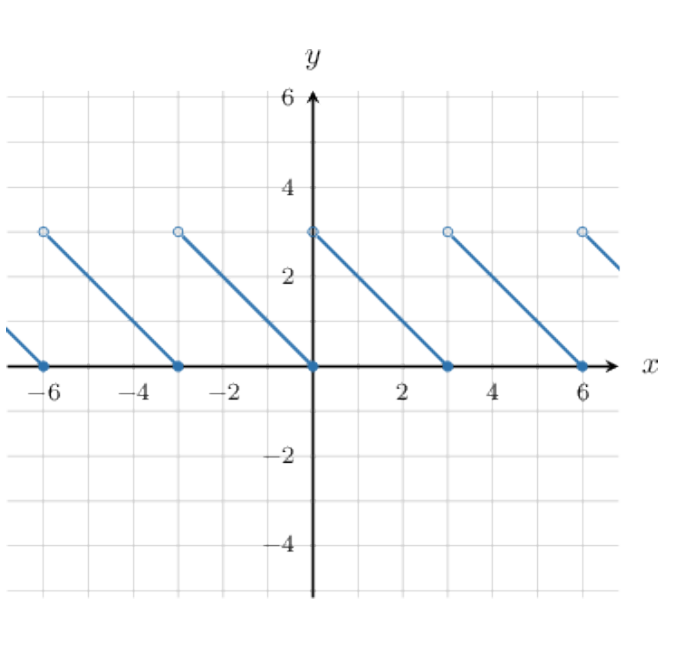
$$\begin{aligned} n = 1: &\quad f(4 + 2(1)) = f(6) = 0 \\ n = 2: &\quad f(4 + 2(2)) = f(8) = 0 \\ n = 3: &\quad f(4 + 2(3)) = f(10) = 0 \\ &\quad \vdots \end{aligned}$$

The same is true for negative values of n. Setting $n = -1, -2, -3 \ldots$ gives the following:

$$\begin{aligned} n = -1: &\quad f(4 + 2(-1)) = f(2) = 0 \\ n = -2: &\quad f(4 + 2(-2)) = f(0) = 0 \\ n = -3: &\quad f(4 + 2(-3)) = f(-2) = 0 \\ &\quad \vdots \end{aligned}$$

Example: Identifying the Period of a Function Given Its Graph

What is the period of the function shown below?



EXPLANATION

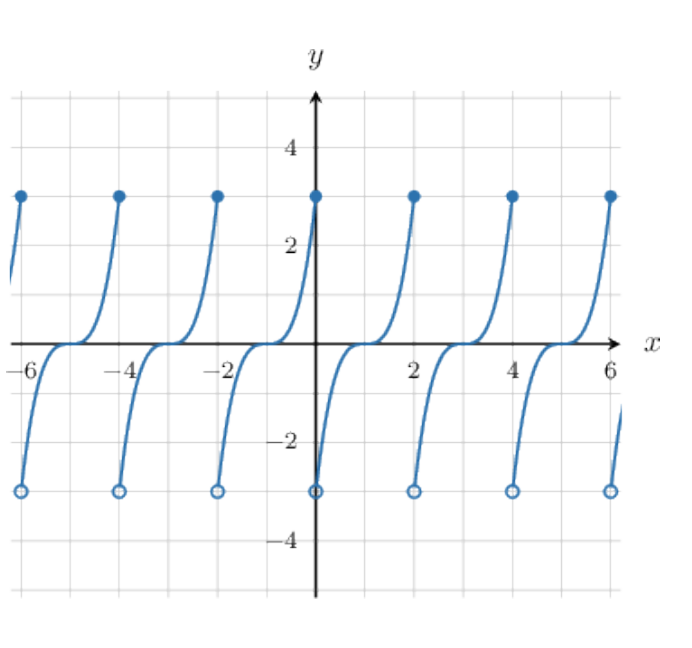
Note that if we shift the graph 3 units left or right, the graph remains unchanged.

On the other hand, if we shift the graph horizontally by any distance other than 3 (or a multiple of 3), we get a different graph.

Therefore, the period of the function is 3.

KEY PREREQUISITE
Piecewise Functions

Question 1



The graph of a periodic function $y = f(x)$ is shown above. What is the period of this function?

- a $\frac{1}{4}$
- b $\frac{1}{2}$
- c 2
- d 3
- e 1

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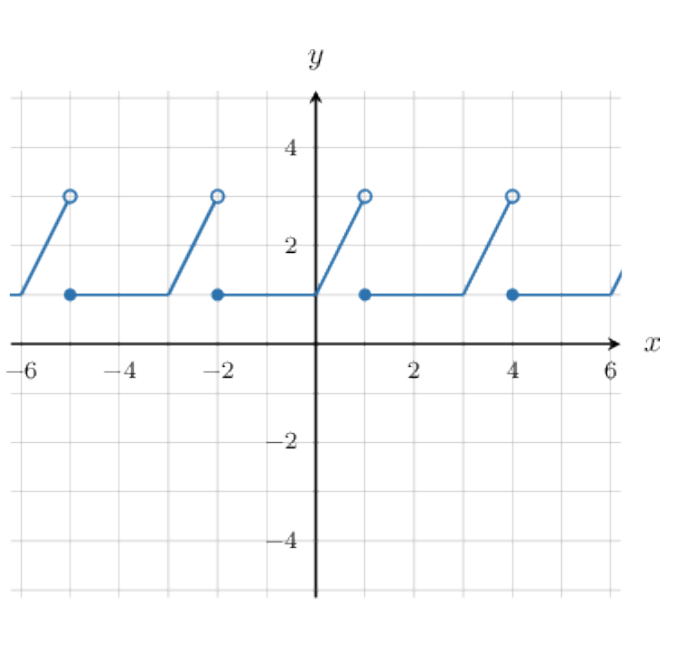
Question 2



The graph of a periodic function $y = f(x)$ is shown above. What is the period of this function?

- a 4
- b 3
- c 2
- d 5
- e 1

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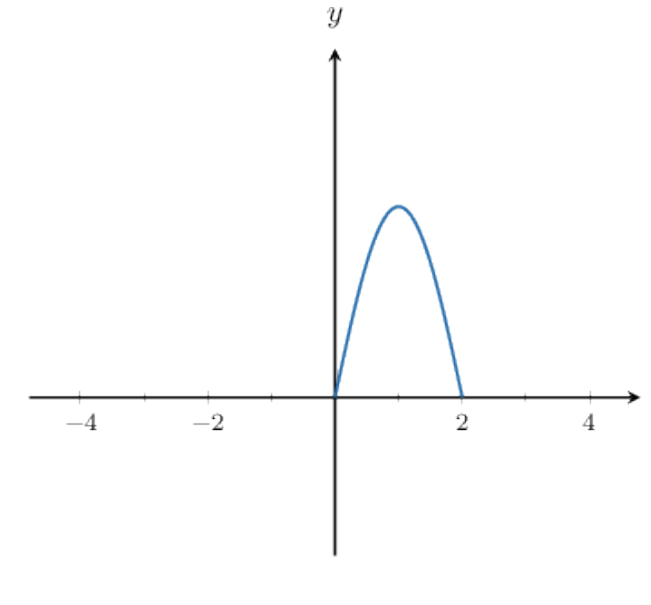
Example: Drawing the Graph of a Periodic Function

Part of the graph of a function $y = f(x)$ with period 2 is shown below. What is the graph of $y = f(x)$ on the interval $-4 \le x \le 4$?

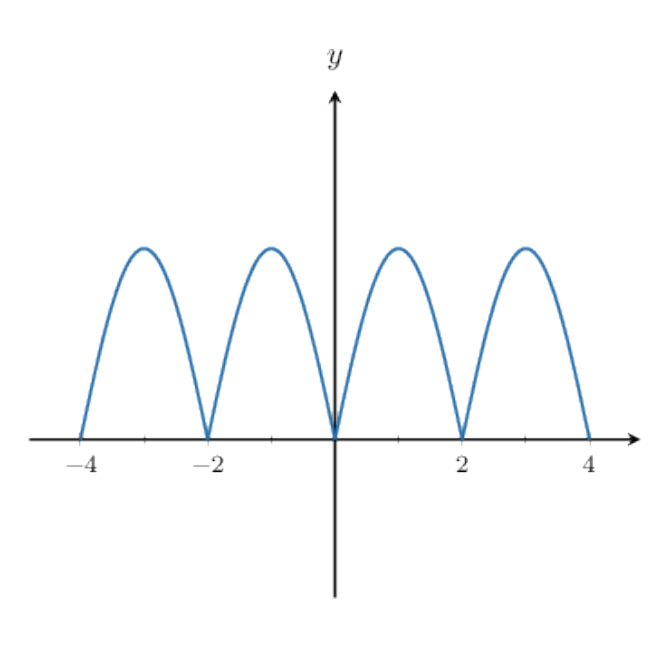


EXPLANATION

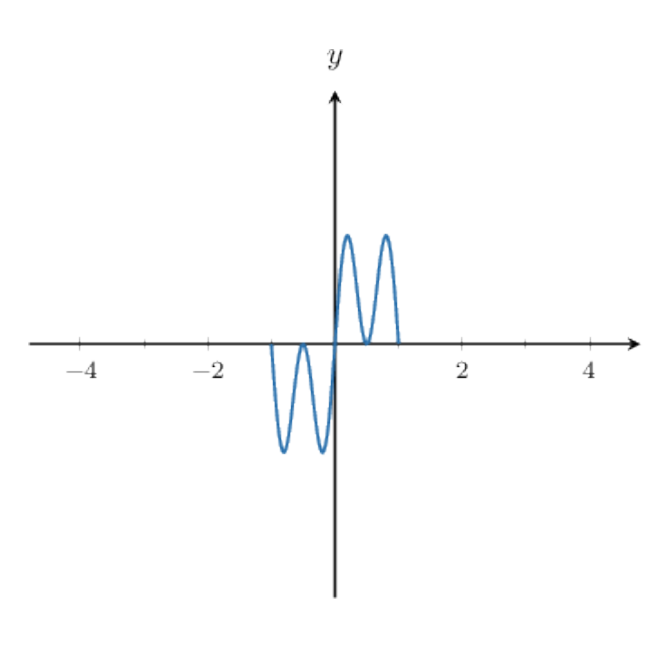
The graph above shows the function in the interval $0 \le x \le 2$.

Since this interval has length 2, and the period of the function is also 2, this interval represents a single period of the function.

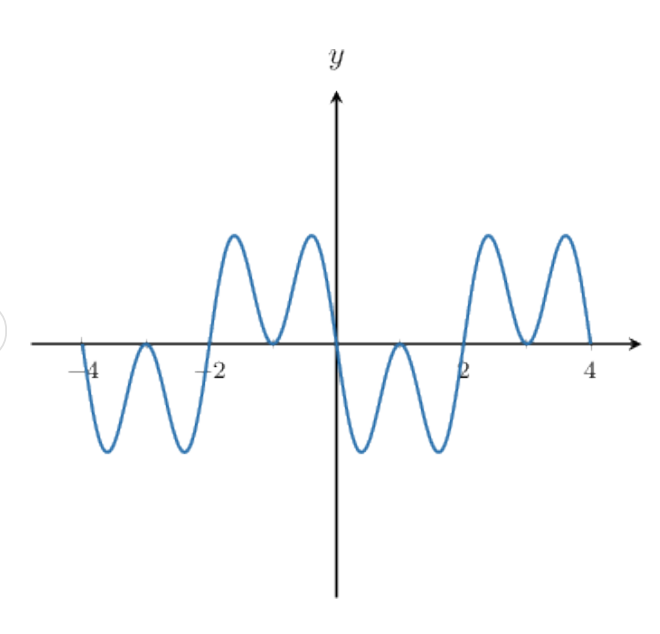
Therefore, the function will consist of many copies of the shape shown above, shifted horizontally by multiples of 2.

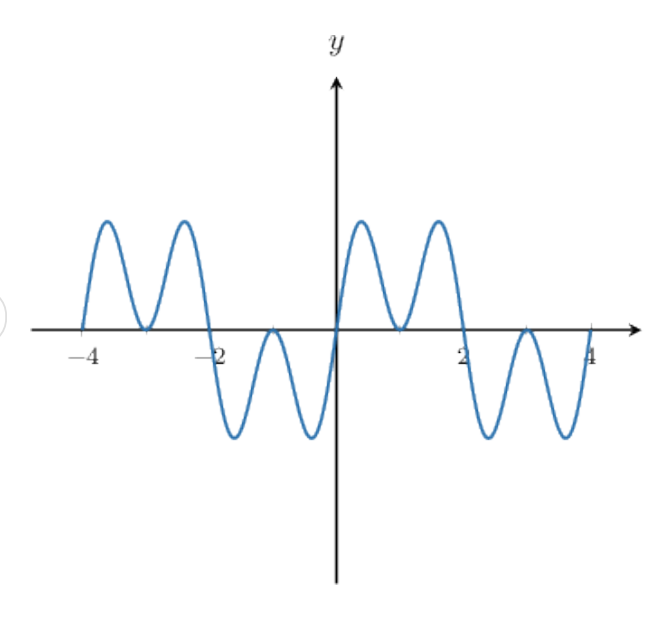


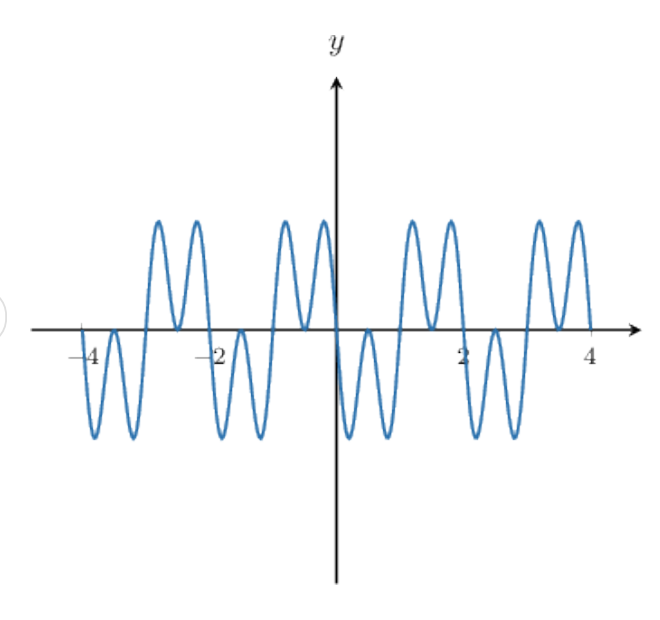
Question 3

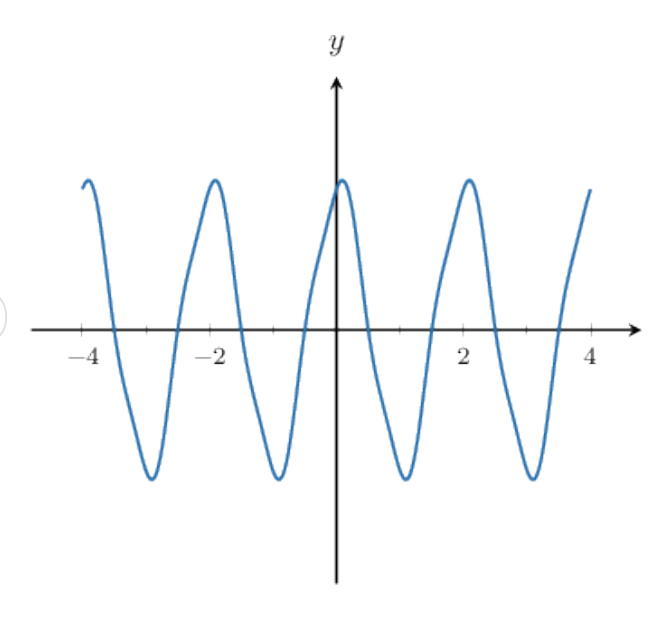


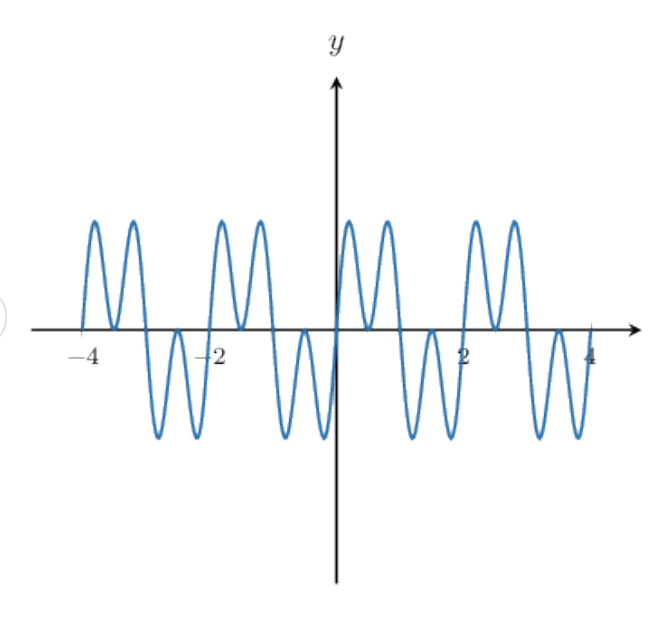
Part of the graph of a function $y = f(x)$ with period 2 is shown above. What is the graph of $y = f(x)$ on the interval $-4 \le x \le 4$?

a


b


c


d


e


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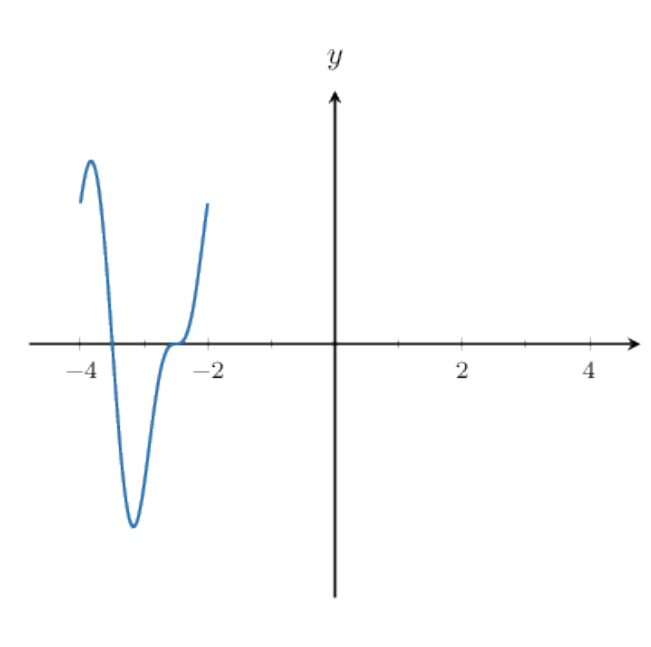
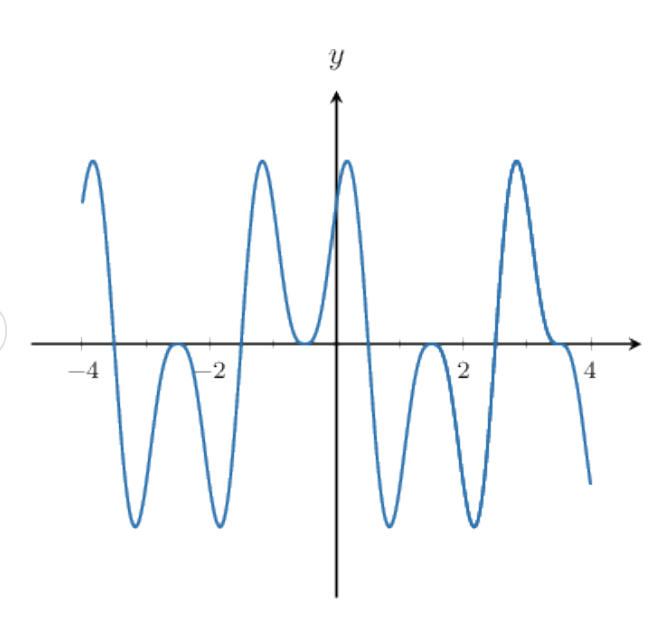
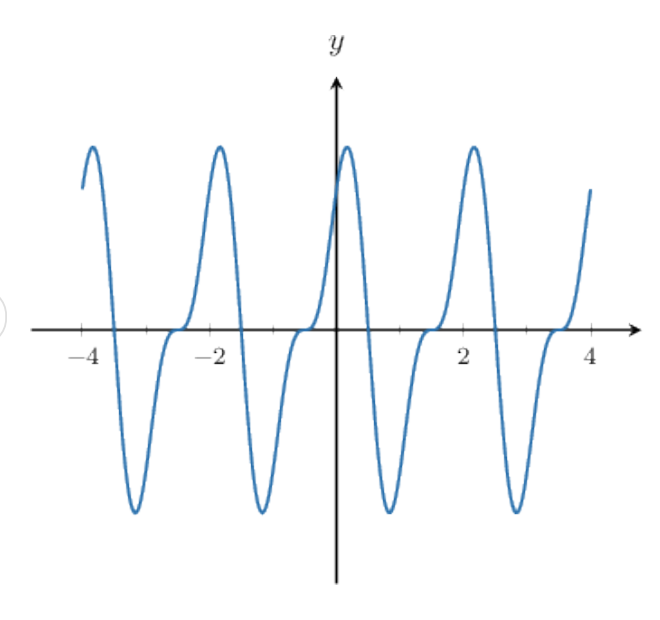
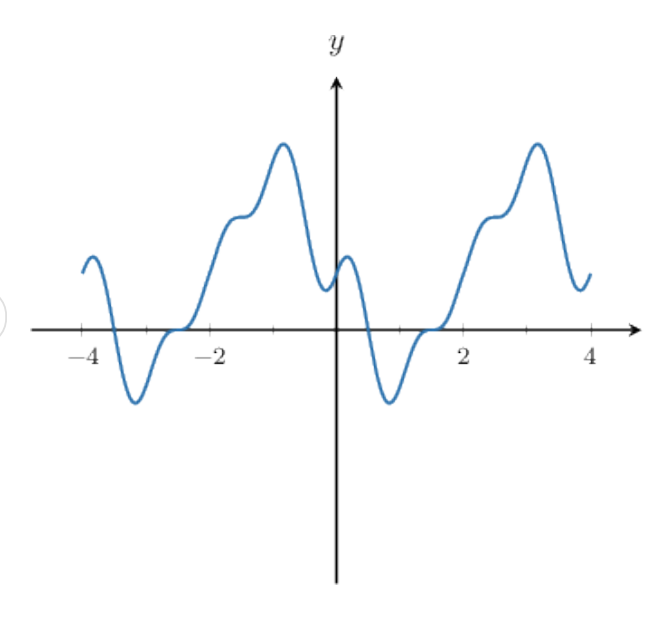
Question 4

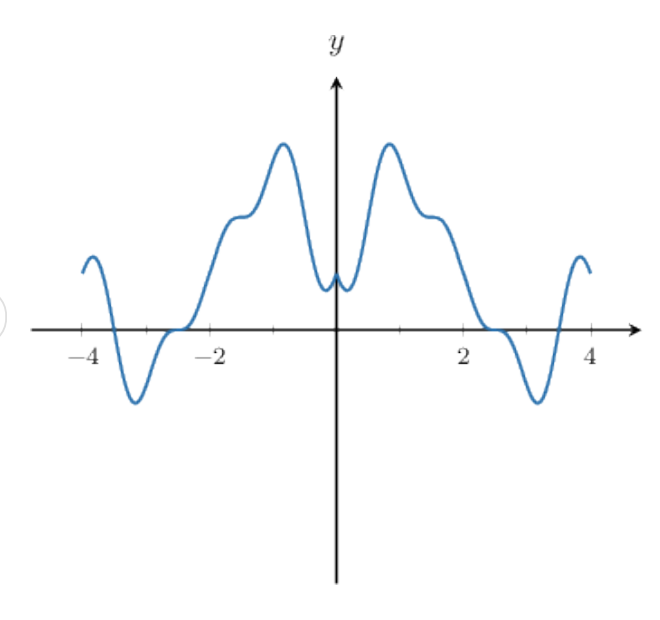


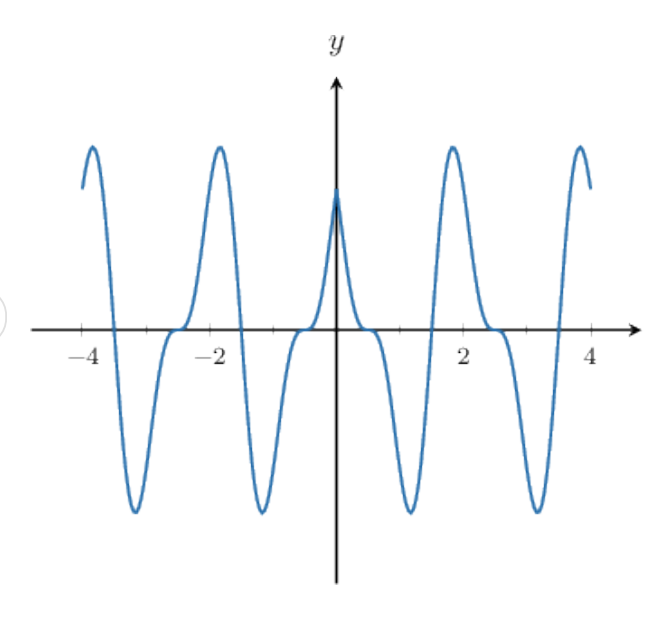
Part of the graph of a function $y = f(x)$ with period 2 is shown above. What is the graph of $y = f(x)$ on the interval $-4 \le x \le 4$?

a


b


c


d


e


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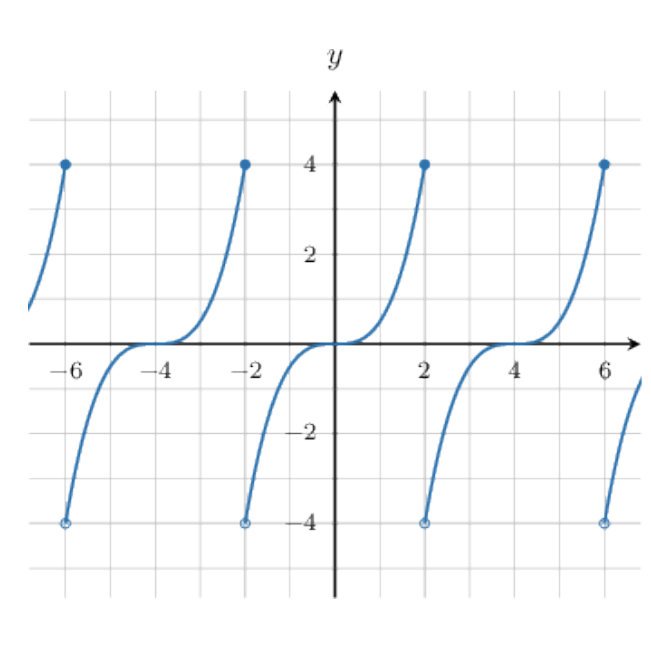
Example: Inferring Values of a Periodic Function Given Its Graph

The graph of $y = f(x)$ is shown below. Given that f is periodic with period 4, what is the value of $f(84)$?



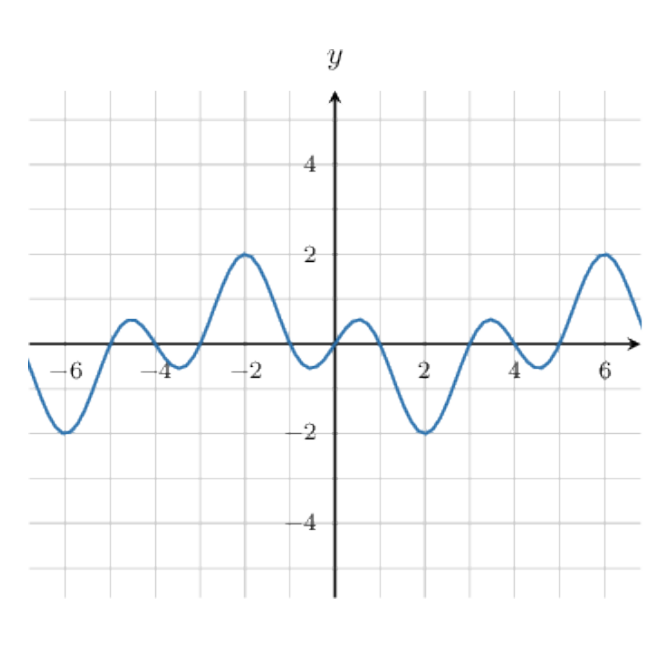
EXPLANATION

Since the period is 4, if we add (or subtract) any multiple of 4 from the input, the output stays the same.

So, we can keep subtracting 4 from the input until we get to an input that we see in the graph:

$$\begin{aligned} f(84) &= f(84 - 4) \\ &= f(84 - 2 \cdot 4) \\ &= \cdots \\ &= f(84 - 20 \cdot 4) \\ &= f(4) \\ &= 0 \end{aligned}$$

Question 5



The graph of $y = f(x)$ is shown above. Given that f is periodic with period 8, what is the value of $f(111)$?

- a -2
- b 2
- c -1
- d 1
- e 0

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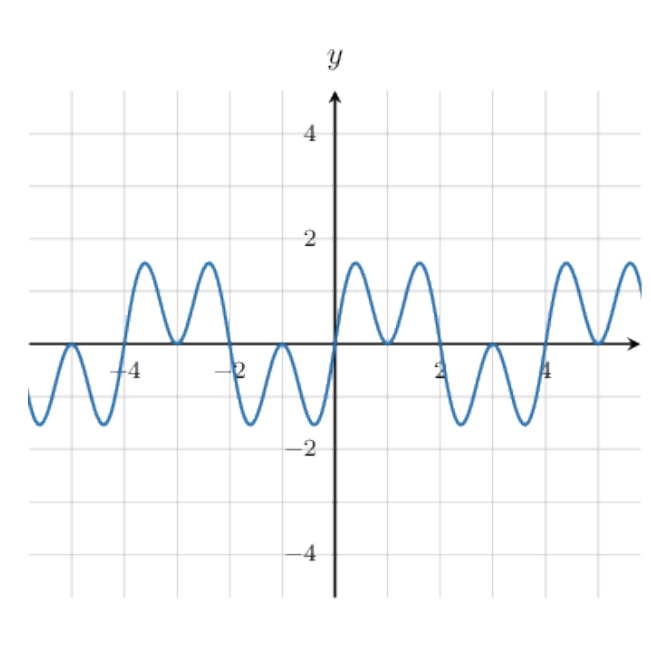
Question 6



The graph of $y = f(x)$ is shown above. Given that f is periodic with period 4, what is the value of $f(42)$?

- a -2
- b -1
- c 2
- d 1
- e 0

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